Renjie Zhao

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Education

University of Washington September 2023 – Expect June 2027
Bachelor of Science in Computer Science (Data Science option) *Seattle, WA*

Work Experience

University of Washington, RealityLab June 2025 – February 2026
Research Assistant

Renjie Zhao, Jingwei Ma, Huy Huynh, Brian Curless, Steven M. Seitz, Ira Kemelmacher-Shlizerman. "GarmentZoom: Generating Zoomable Images from Garment Listings", ECCV 2026 RetailVision Workshop. [Project Page] [arXiv]

- Built world's **first** generalized super resolution model capable of 3–20 super resolution for garment images and online listings.
- **Fine-tuned** Flux-1.0-dev to accept extra control of reference texture image.
- Built the data curation pipeline to form non-aligned training pairs.

University of Washington, CSE 453/M553: Data Center Systems September 2026 – December 2026
Teaching Assistant

- Lead class discussion sections.
- Design and test class coding projects used by 133 students.

Research Experience

Allen Institute for AI (Ai2) February 2026 – Present
Student Researcher (Project Lead) | *Improving Physical Realism in Video Generation Models through Reinforcement Learning*

- Training a video diffusion model (Wan-2.2 27B-A14B) with large scale **reinforcement learning** to improve physical realism in generated videos.
- Optimized **16-GPU (Nvidia H200 141GB)** RL training, achieving 4× faster steps than Flow-GRPO.
- Designed novel **data retrieval pipeline** for large scale dataset curation.
- Developed an **automated reward** for reinforcement learning.

UW SysLab December 2024 – June 2025
Student Researcher | *Power Adaptive Storage System with Zoned Storage Devices*

- Modified F2FS **File System** in **Linux kernel** to utilize the predictable power profiles of zone-namespace storage devices.
- Stabilized IO performance with background garbage collection and data migration.

Projects

ML-TOMB: Multi-LoRA Training with One-Model Base [Report]
Renjie Zhao, Alek Curless

- Implemented a **CUDA kernel** for training multiple LoRA adapters on a shared base model.
- Achieved up to 2.63× speedup for machine learning training.
- Proposed a novel framework to **benchmark** training speed.

SiriTok - Semantically Rich 1D Tokenizer [Poster]
Renjie Zhao

- A one-dimensional image tokenizer for latent diffusion models while preserving both textural details and semantic information.
- Achieved 79% linear probing **classification accuracy** on ImageNet from our latent space.
- Proposed a novel semantic preservation loss.
- Fine-tuned a pretrained semantic encoding model (DINOv3) with **8-GPU** (Nvidia A40 48GB) FSDP.

Technical Skills

Languages: Python, C/C++, CUDA, Java, C#, JavaScript, SQL, GoLang
ML / Systems: PyTorch, MuJoCo, Isaac Sim, ROS, NumPy
Areas: Computer Vision, Machine Learning, ML Systems, Reinforcement Learning, Distributed Systems